University of the West Indies, Cave Hill Campus
Department of Computer Science, Math and Physics

LAB 1 - Industrial Node Power Supply

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Introduction

Modern industrial control systems rely on distributed electronic nodes to monitor, process, and control physical processes. Each of these nodes requires a stable and reliable power supply in order to operate sensors, actuators, and control electronics safely and accurately. Poor power regulation can lead to measurement errors, malfunction of control devices, and system instability. For this reason, the design of industrial power supply is a fundamental skill in automation, instrumentation, and control systems.

This laboratory exercise introduces the fundamentals of a basic industrial control node by focusing on the design and implementation of a regulated DC power supply capable of powering external I/O devices. The power supply must convert an AC input source into a clean, stable DC output with minimal ripple and good voltage regulation, even when the input voltage or load current changes.

Throughout this lab, the principles of rectification, filtering, and voltage regulation will be applied, and will gain practical experience in analyzing regulator performance under varying operating conditions.

Section 1 - Design and Simulation

Block Diagram

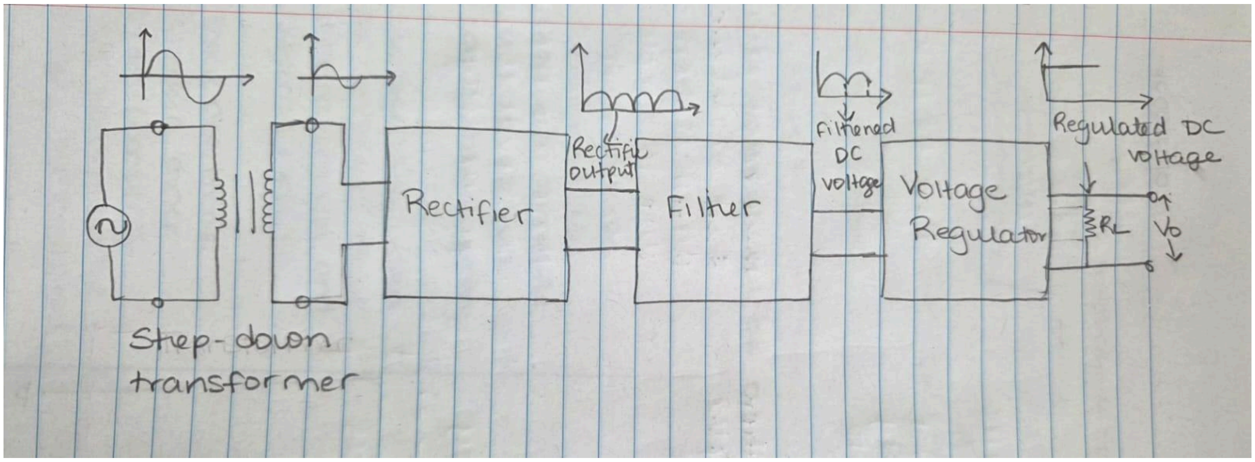

Figure 1: Block Diagram of Industrial node

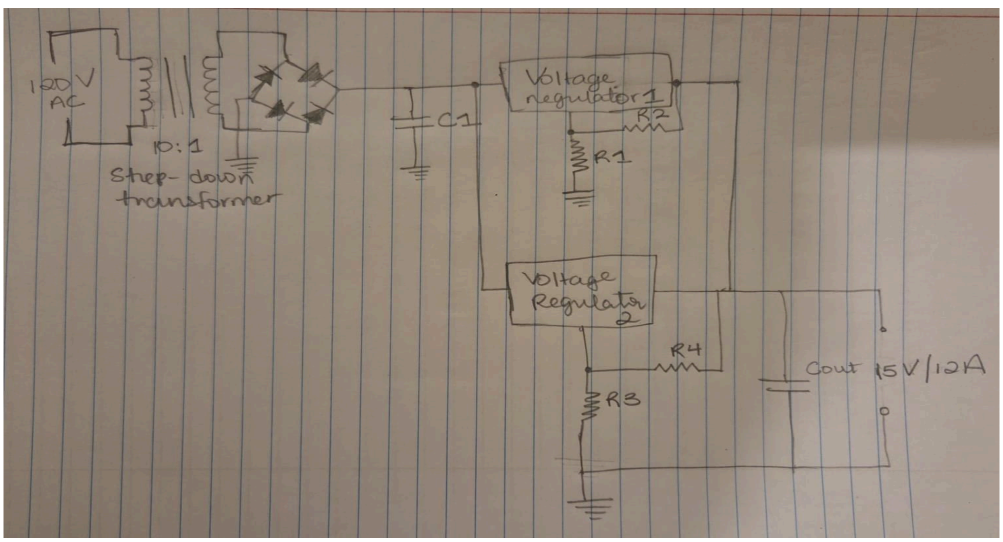

Figure 2: Design of Power Supply

Theoretical Review of Circuit

Step-Down Transformer

The transformer converts the 120V AC supply into a lower AC voltage suitable for the DC power supply. It also provides electrical isolation, which is essential for safety in industrial systems. The secondary voltage is selected so that, after rectification and filtering, there is sufficient voltage headroom for regulation.

Bridge Rectifier

The bridge rectifier converts the transformer's AC output into a full-wave pulsating DC. During each half-cycle, two diodes conduct, ensuring that current always flows in the same direction through the load. This stage provides a unidirectional voltage that can be filtered and regulated.

Input Smoothing Capacitor

The large electrolytic capacitor placed after the rectifier smooths the pulsating DC by storing energy during the voltage peaks and releasing it during the valleys. This reduces ripples and provides a stable DC input to the voltage regulators.

Adjustable Voltage Regulator

Two linear voltage regulators are connected in parallel. Each regulator attempts to maintain the same output voltage. By operating in parallel, the total load current is shared between them, increasing the maximum current the supply can deliver. Small equalising resistors at the outputs ensure that the current is shared evenly between the regulators and prevent one from supplying most of the current. This prevents the regulators from overheating and improves stability.
Each regulator continuously compares its output voltage to its internal interface. If the output voltage drops due to increased load, the regulators increase their output current. If the voltage rises, they reduce their output. Since both regulators are set to the same output voltage, they work together to maintain constant DC output.

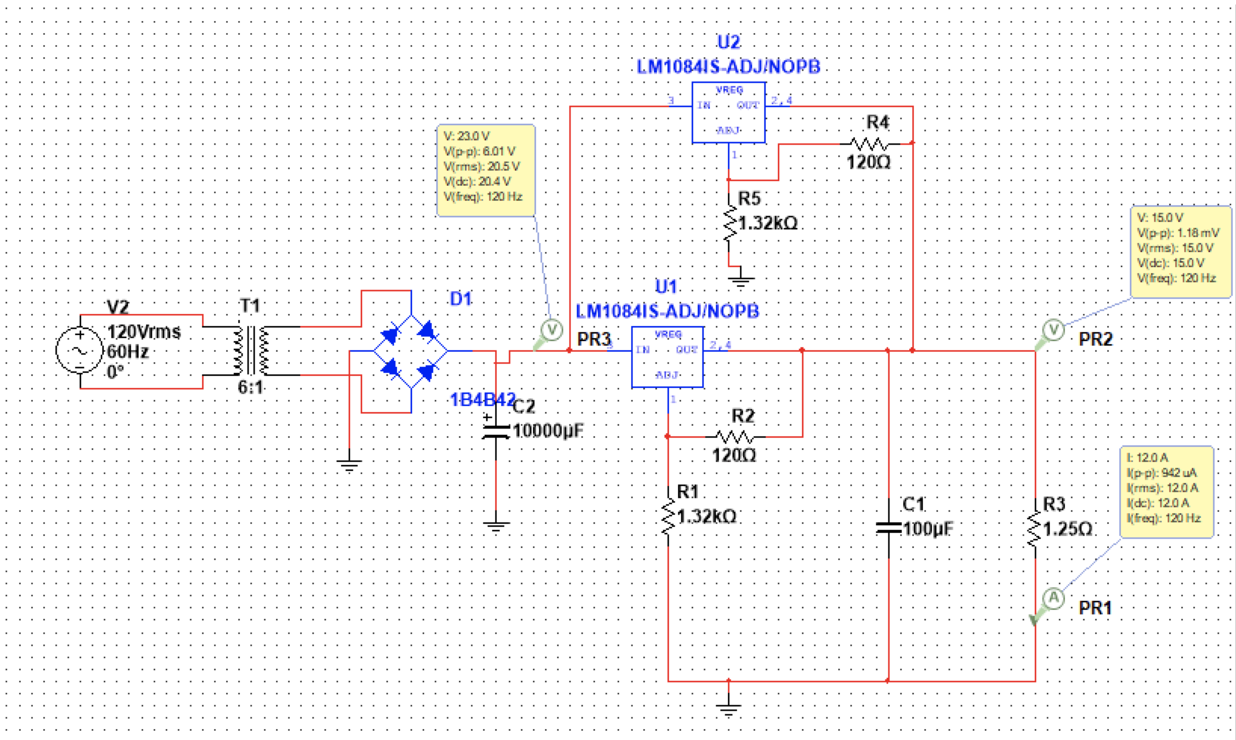

Figure 3: Simulation of Power Supply Circuit in Multisim

Input Voltage below (V)	Output Voltage (V)	Input Voltages above (V)	Output (V)
11	14.7	23	15V
14.3	14.7	38.4	15V

Section 2 - Practical Testing

Components Used:

- Bread Board
- Wires
- 1k Ohm & 329 Ohm resistors
- 1 47uF capacitor
- 4 1N4007 diodes
- 12V AC supply
- LM317K Regulator
- Multimeter

Simulated AC value	Measured AC value
20.4V	15V

Simulated R1	Measured R1	Theoretical R1
1k Ohms	1.005k Ohms	1k Ohms

Simulated R2	Measured R2	Theoretical R2
330 Ohms	329 Ohms	330 Ohms

Simulated Vout	Measured Vout	Theoretical Vout
5.10 V	5.11V	5.04V

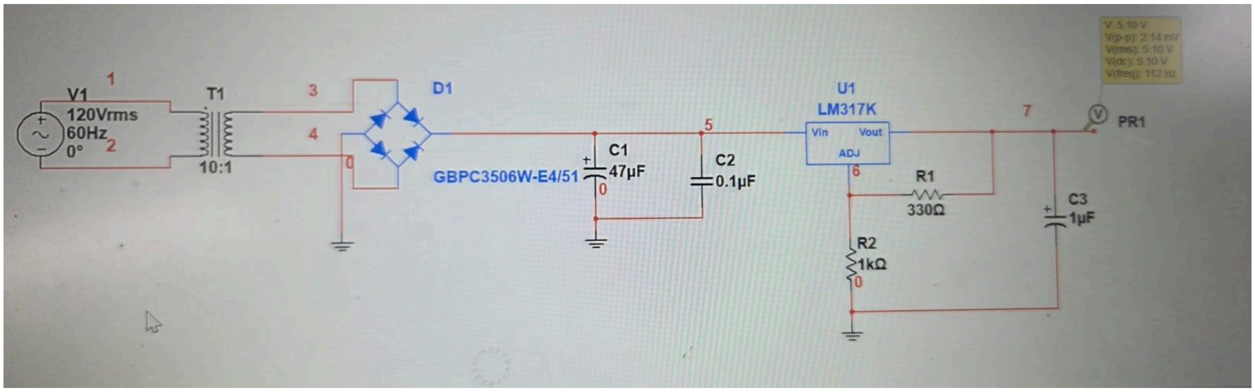

Figure 4: Simulated Circuit in Multisim

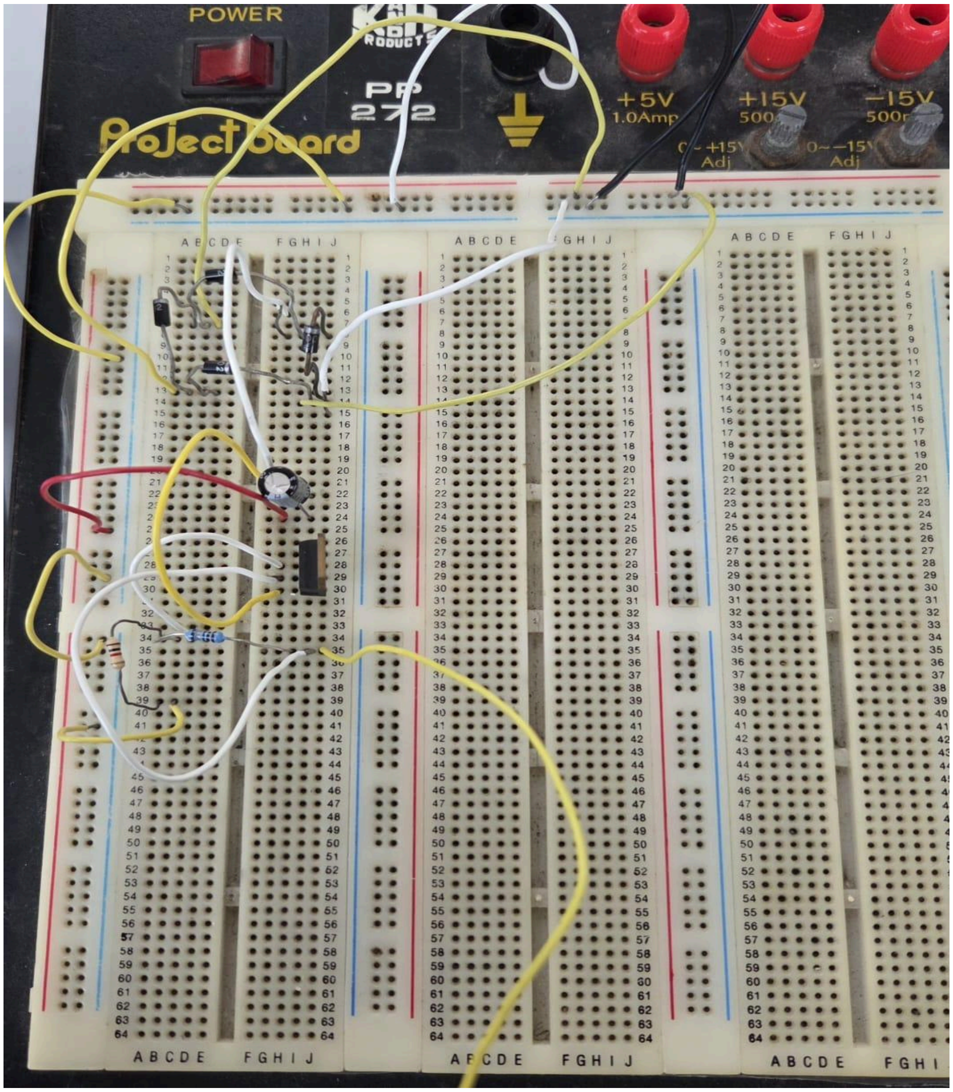

Figure 5: Practical Circuit on Bread Board

Results

Calculation for Components Chosen

$R_1 = 1k\Omega$, $R_2 = 330\Omega$

Vout according to datasheet:

$$V_{out} = 1.25V\left(1 + \frac{R_1}{R_2}\right)$$

$$= 1.25\left(1 + \frac{1000}{330}\right)$$

$$= 1.25(4.03) = \underline{\underline{5.04V}}$$

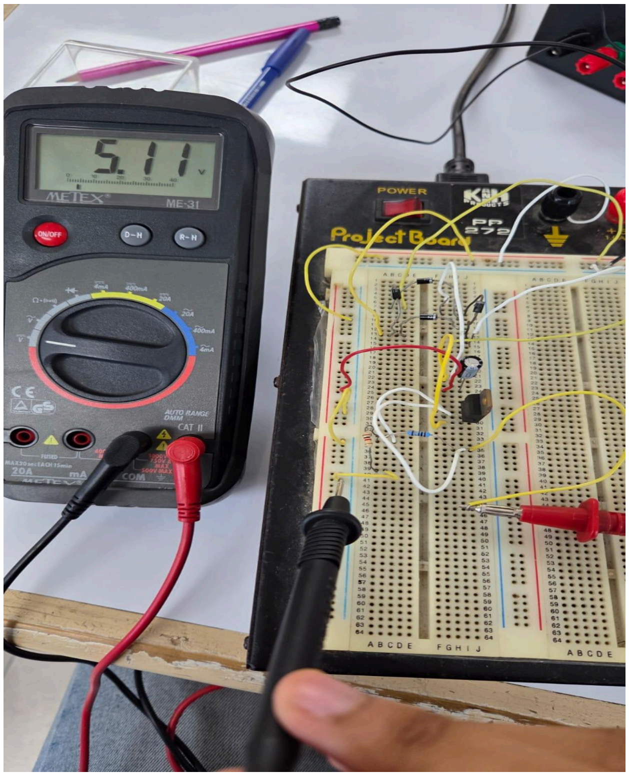

Figure 6: Circuit showing measured output

Discussion

The objective of this laboratory was to design and evaluate a regulated DC power supply suitable for powering an industrial control node. The circuit was developed using a step-down transformer, a full-wave bridge rectifier, a smoothing network, and an adjustable linear voltage regulator (LM317K) to obtain a stable 5.10V DC output from a 120V AC mains source.

The transformer reduced the 120V RMS input to approximately 20.4V, which is a safe and appropriate level for rectification and regulation. After passing through the bridge rectifier, the AC waveform converted into pulsating DC. The peak rectified voltage was approximately 16-17V, which provided sufficient headroom above the desired 5V output for the LM317 regulator to operate correctly.

The filter capacitor C1 significantly reduced the ripple in the rectified voltage by storing charge at the peaks of the waveform and releasing it during the valleys. This produced a smoother DC voltage at the input of the regulator. The small bypass capacitor C2 further suppressed high-frequency noise and improved regulator stability, which is essential in industrial systems where electromagnetic interference may be present.

The LM317K adjustable regulator was configured using resistors R1 and R2 to produce an output voltage of approximately 5.10V. The measured voltage confirms that the regulator was operating correctly and maintaining a constant voltage despite ripple on the input. This demonstrates the effectiveness of the regulator in isolating the load from variations in the supply voltage. The output capacitor C3 improved transient response and further reduced output ripple, resulting in a clean DC output suitable for powering sensitive electronics.

Overall, the results demonstrate that the designed circuit successfully converts high-voltage AC into a low-voltage, stable DC supply. The system meets the requirements of an industrial node power supply. This confirms the design approach is suitable for powering industrial control and monitoring equipment.